Narrow structures: count-error improvement does not establish parameter recovery

All nine designs with defined location / width truth | widths and parameter errors in distance-modulus mag

No fictitious structure truth is assigned to the other 13 scenarios. Right labels give all / accepted catalog denominators.

All eight catalogs, failures retained

Local arm numerically accepted

Structure location | Bias

narrow 4 kpc
narrow 16 kpc w0.08
narrow 16 kpc w0.25
narrow 2 kpc w0.08
narrow 2 kpc w0.25
narrow 4 kpc w0.08
narrow 4 kpc w0.25
narrow 8 kpc w0.08
narrow 8 kpc w0.25

−0.14 −0.12 −0.10 −0.08 −0.06 −0.04 −0.02 0.00

Mean posterior-median error (mag)

Structure location | Coverage

n=8/8
n=8/7
n=8/7
n=8/8
n=8/8
n=8/8
n=8/8
n=8/7
n=8/8

0 20 40 60 80 100

Empirical 95% interval coverage (%)

Structure width | Bias

narrow 4 kpc
narrow 16 kpc w0.08
narrow 16 kpc w0.25
narrow 2 kpc w0.08
narrow 2 kpc w0.25
narrow 4 kpc w0.08
narrow 4 kpc w0.25
narrow 8 kpc w0.08
narrow 8 kpc w0.25

0.00 0.05 0.10 0.15 0.20

Mean posterior-median error (mag)

Structure width | Coverage

n=8/8
n=8/7
n=8/7
n=8/8
n=8/8
n=8/8
n=8/8
n=8/7
n=8/8

0 20 40 60 80 100

Empirical 95% interval coverage (%)

Intervals resample whole fields conditional on the shared source bank. Source-overlap blocks collapse to one for these designs: no source-level interval.

The 95% line is nominal; eight correlated simulation realizations cannot establish independently calibrated real-population coverage.

All numerical failures and the original entry-table local-dust approximation remain in the saved products and this comparison.